

ANTENNA - DETAIL SPECIFICATION

two required - row N, gores 18 and 4 - 3.75 m overall

OVERALL

Height	3,750 mm = 2,250 mm cone (axial) + 1,500 mm ball
Attitude	leans 35 deg outboard from the LOCAL SURFACE NORMAL at row N
Inflation	ram-air: fed from inside the envelope through the panel-N feed hole
Colours	cone black ripstop, ball red ripstop (#e9132f), doubler green (it is another panel N)

BALL

Diameter	1,500 mm
Gores	16 per ball (32 total) - see the note on gore count below
Gore size	2124 mm long x 294.5 mm at the equator (finished)
Neck opening	O312 mm - gores truncated at 12.0 deg from the pole
Top cap	O150 mm disc - gores truncated at 174.3 deg; avoids a 16-seam pile-up
Equator	at 1021 mm up each gore - match-mark every gore
Zip	350 mm, in ONE meridian seam, centred on the equator
Faceting	14.4 mm sagitta = 1.9 % of radius (12 gores would be 26 mm, 20 gores 9 mm)

CONE

Size	O875 mm base to O312 mm neck, 2250 mm axial, 2268 mm slant
Panels	2 per cone (4 total) - see the fabric-width note below
Development	radius 1257 to 3524 mm nominal, 44.69 deg included
Base edge	OBLIQUE - developed radius runs 3240 to 3863 mm; this is what creates the lean
Seams	2 meridian seams: one OUTBOARD (long side), one INBOARD (short side)

ATTACHMENT TO THE ENVELOPE

Opening in skin	ELLIPSE 1076 mm (fore-aft) x 878 mm (across) - NOT a circle
Base reinforcement	a SECOND PANEL N on gores 18 and 4, sewn into all four seams - not a floating patch
Why	the O575 feed hole removes 11 % of panel N; a free O1250 disc would leave only 36 mm to the row fells
Feed hole	O575 cut / O625 finished, inside the ellipse, centred on the ellipse centre
Orientation	ellipse LONG axis runs radially outboard - marked on the opening template
Webbing	Class 3 1/2 in (12.7 mm) load tape: base ellipse 3.08 m + feed-hole hem 1.96 m per antenna; neck ring 1.0 m
Base edge	the cone's oblique base edge measures 3.08 m, NOT the 7.90 m of a straight cut

MATERIALS PER ANTENNA (corrected)

Black ripstop	cone 4.23 m2 per antenna (8.5 m2 for both)
Green ripstop	2 extra panel N (2.34 m2 each, 4.68 m2 total) - cut them with the other panels
Red ripstop	ball 7.07 m2 (+10 % nesting waste = ~7.8 m2)
Webbing	25 mm: 4.0 + 2.8 + 1.0 = 7.8 m
Thread / tools	V-69 (or V-92 at the base ring); hot knife for every edge
Hardware	350 mm coil zip; optional LED cluster inside the ball

Previous sheets stated 2.2 m2 black and 1.9 m2 red - both were far too low; the figures above are the true developed areas.

[Sheets 2-3 of this document: sewing sequence.](#) [Cutting templates at 1:1 are in alien-balloon-antennae.pdf.](#)

ANTENNA - SEWING THE BALL

O1,500 mm, 16 gores, red ripstop - build both balls together so they match

- 1

CUT

Cut 16 gores per ball (32 total) from the sheet-4 template, hot knife. Cut 2 top caps (O150 finished). Mark the EQUATOR notch on every gore edge before moving them.
- 2

PAIR UP

Sew gores into 8 pairs. 25 mm French fell, equator notches matched. Sew from the equator OUT to each pole - never pole-to-pole, or the halves creep and the ball goes lopsided.
- 3

QUARTERS

Join pairs into 4 sets of four, then into 2 half-balls. Match equator notches at every seam.
Check each half lies flat as a semicircle before going on.
- 4

ZIP

Fit the 350 mm zip into ONE meridian seam of one half, centred on the equator. Sew the zip tape in with the fell so the seam keeps its strength; bar-tack both ends of the zip.
- 5

CLOSE

Join the two halves: one seam closed normally, the last seam closed with the zip open so you can reach inside. All {NG} seams now meet at each pole.
- 6

NECK

Trim the pole ends true to the O312 mm neck circle. Turn 25 mm and sew the neck webbing ring (1.0 m) inside the turn. This ring is the ball's load path - twin-needle it.
- 7

TOP CAP

Trim the top pole to the O150 mm circle and sew the cap disc over it, turning the disc edge under. Twin-stitch round twice. The cap hides the 16-seam junction and stops it leaking.
- 8

CHECK

Inflate with a leaf blower through the neck. The ball should be round to about 15 mm; a flat facet means one seam is tight - release and re-sew that seam rather than forcing it.

Equator sits 1021 mm up from the neck end of each gore - the template carries that notch.

Sew both balls before starting the cones: it is much easier to match a pair while the jigs are set.

ANTENNA - SEWING THE CONE AND FITTING IT TO THE ENVELOPE

black ripstop - 2 panels per cone - the base edge is CUT OBLIQUE, this is what leans the antenna

- 1

CUT

Cut 2 panels per cone (4 total) from the CONE PANEL sheet in alien-balloon-antennae.pdf. The two long edges are NOT the same length: the OUTBOARD edge is 3863 mm and the INBOARD edge 3240 mm from the neck arc. Mark them.
- 2

SEAMS

Join the panels into a cone with two meridian seams, 25 mm French fell. Match the printed match-marks. Keep both seams the same tension or the cone will spiral.
- 3

BASE RING

Turn the oblique base edge 25 mm and sew 3.08 m of Class 3 1/2 in tape into the turn, twin-needle. This ring carries the whole antenna load into the doubled panel.
- 4

NECK RING

Turn the neck edge and sew the 1.0 m webbing ring in. Then join cone neck to ball neck, ring to ring, O312 mm. Bar-tack at four quarter points.
- 5

DOUBLE THE PANEL

Cut 2 extra panel N from the normal panel template. Lay one on the INSIDE face of panel N on gore 18 and one on gore 4, edges flush, and treat the pair as a single ply from here on - they are caught into the M/N and N/O row fells and both gore fells when the envelope is assembled.
- 6

FEED HOLE

On panel N (gores 18 and 4) cut the O575 mm feed hole and double-fold it to O625 mm finished. It sits inside the ellipse, on the ellipse centre.
- 7

MARK THE OPENING

Lay the sheet-6 template on the doubled panel with the ellipse LONG AXIS pointing radially OUTBOARD (away from the crown) - the template marks this. Mark the ellipse, then cut and hem the O575 feed hole through BOTH plies together, hemmed over one layer of Class 3 1/2 in tape and double-folded to O625. Bar-tack at four points. just inside the webbing ring.
- 8

JOIN CONE

Sew the cone base ring to the marked ellipse, matching the OUTBOARD mark on the cone to the OUTBOARD mark on the panel. Twin-needle plus bar-tacks every 150 mm, through both plies.
- 9

TEST

Ground-inflate the envelope. The antenna should stand up on its own at roughly 35 deg outboard. If it flops, the base ring is loose; if it stands upright, the oblique edge has been fitted the wrong way round.

Fit the antennae AFTER the gores are joined but BEFORE the parachute is rigged - you need to reach inside panel N.

Load path: ball neck ring -> cone neck ring -> cone base ring (Class 3 1/2 in tape) -> DOUBLED panel N -> the four seams round it and their load tapes.